

Energy efficiency drives the global seasonal distribution of birds

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The uneven distribution of biodiversity on Earth is one of the most general and puzzling patterns in ecology. Many hypotheses have been proposed to explain it, based on evolutionary processes or on constraints related to geography and energy. However, previous studies investigating these hypotheses have been largely descriptive due to the logistical difficulties of conducting controlled experiments on such large geographical scales. Here, we use bird migration—the seasonal redistribution of approximately 15% of bird species across the world—as a natural experiment for testing the species–energy relationship, the hypothesis that animal diversity is driven by energetic constraints. We develop a mechanistic model of bird distributions across the world, and across seasons, based on simple ecological and energetic principles. Using this model, we show that bird species distributions optimize the balance between energy acquisition and energy expenditure while taking into account competition with other species. These findings support, and provide a mechanistic explanation for, the species–energy relationship. The findings also provide a general explanation of migration as a mechanism that allows birds to optimize their energy budget in the face of seasonality and competition. Finally, our mechanistic model provides a tool for predicting how ecosystems will respond to global anthropogenic change.

One of the most promising hypotheses proposed to explain the global distribution of biodiversity is the species–energy relationship, which states that the constraints of energy supply and demand drive animal diversity^{1,2}. In particular, according to Lotka's maximum power principle³, natural selection should favour organisms that have the most advantageous balance between the energetic costs associated with their environment and behaviour and the benefits in terms of access to available energy to fuel metabolism⁴. Accordingly, energy efficiency has been shown to explain several species' traits such as body size⁴ and clutch size⁵. Energy efficiency has also been shown to affect the movement behaviour and spatial distribution of species as they optimize energy acquisition^{6,7}. Species, however, do not exist in isolation, and the energy efficiency of a particular distribution depends not only on the quantity of available resources but also on the distribution of the other species competing for the same resources. Overall, this should translate into an energy-efficient distribution of all species across the world, in which each species minimizes its energy expenditure while targeting areas with maximum energy available given the distribution of the other species. Macroecology has provided support for this hypothesis, with studies revealing a positive and linear relationship between species richness and the amount of available energy in the environment^{8,9}. However, these studies were mainly descriptive (for example, ref. ¹⁰) and a mechanistic understanding of this relationship has thus far proven elusive.

Birds are highly mobile organisms and, twice a year, approximately 15% of species migrate between breeding and non-breeding grounds¹¹ in a “seasonal ecological adjustment on a gigantic scale”¹². Bird migration therefore provides a natural experiment for testing hypotheses about the mechanisms driving the spatial distribution of species^{13,14}, as any such mechanisms must explain not only spatial patterns of bird diversity but also the seasonal redistribution of such diversity. Previous macroecological studies of bird migration

have supported the hypothesis that energy drives the seasonal distribution of migratory species^{15,16}. Negative costs are associated with winter harshness and longer migratory distances, whereas benefits are associated with access to seasonally available resources¹⁶. These studies, however, only considered migratory species, were correlative and integrated energy implicitly rather than explicitly. Here, we present a mechanistic model of the geographical distribution of all terrestrial bird species across the world and throughout the year. The model explicitly integrates energy by expressing costs and benefits into a common energetic currency and by accounting for interspecific competition. Built from first ecological and energetic principles, our spatially explicit model simulates a virtual world in which species are distributed in an energy-efficient way. By comparing model outputs to empirical global patterns of bird diversity^{16,17}, we test the prediction that energy efficiency drives the distribution of birds across space and seasons.

Results

Our model was built from the assumptions that the energetic costs associated with any given species' distribution comprise thermoregulation costs, reproduction costs and migration costs (in addition to a basal energy use for existence, which is constant across species). In contrast, energy supply available to a species at any given season is a function of the primary productivity available and not yet appropriated by other species. Each model simulation started with a virtual empty world with the same geography and seasonality as the real world (similar landmass distribution, similar climate, similar primary productivity and with two seasons), which was then progressively filled with virtual bird species. Interspecific variation in morphology, physiology or biology were not incorporated in the model, thus assuming ecological, demographic and energetic equivalence among these virtual species. In each simulation step, a single new species was added, selected as being the most energy

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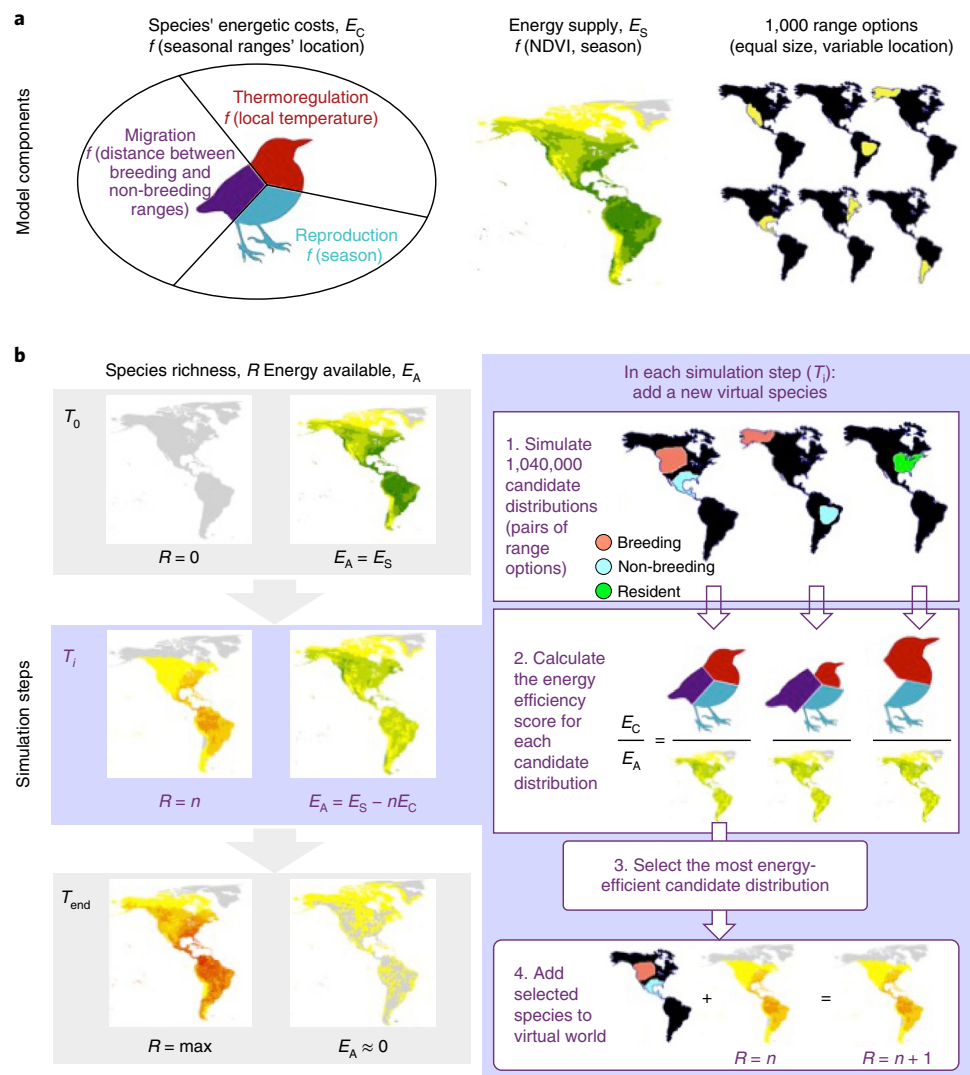


Fig. 1 | Model description. a, The model is built from the following three main components: species' energetic costs (a function of the location of breeding and non-breeding ranges, comprising thermoregulation, reproduction and migration costs); energy supply (derived from the NDVI, and variable across space and seasons); and 1,000 simulated range options (the same size as the average bird seasonal range size). Integrating these three components, the model is applied through a sequence of simulation steps whereby a virtual world with the same geography and seasonality as Earth is progressively filled with virtual species. **b**, At the start of the simulation (T_0), the virtual world is empty of bird species ($R = 0$) and the energy available is equal to the energy supply ($E_A = E_S$). In each simulation step (T_i ; sub-steps 1 to 4) a new virtual species is added to the virtual world, selected among 1,040,000 candidate species (each being a pair of a breeding and a non-breeding range options) by being the most energy-efficient distribution (lowest ratio between energetic costs and the energy available remaining given the n species already present $E_A = E_S - nE_C$). As this new species is added ($R = n + 1$), the energy available E_A is further depleted in the corresponding breeding and non-breeding ranges. The simulation ends (T_{end}) when the virtual world is nearly saturated with simulated species ($E_A \approx 0$ in at least one season). The different size shading of the bird shadows indicate that the corresponding energetic costs (see top-left of the figure) have different values for different candidate distributions. The grey shading indicates that the two maps within it represent the state of the system at a given step. The purple shading highlights that the explanation on the right-hand side represents what happens during one simulation step.

efficient among many candidates (that is, the one whose combination of breeding and non-breeding range results in the most favourable cost–benefit balance), and the energy supply in the virtual world depleted accordingly. New virtual species were added until the virtual world became nearly saturated (see Methods and Fig. 1 for more details). The model included the following four free parameters: three associated with the energetic costs of migration, thermoregulation and reproduction, and one associated with the scaling of energy supply (see Methods and Supplementary Fig. 1). For any combination of parameter values, a virtual world saturated with virtual species was deterministically simulated, from which five geographical patterns representing the global seasonal distribution of birds were derived and contrasted with observations. The

parameter values that best predicted the empirical patterns (that is, the best-fit model) was identified using a genetic algorithm (see Methods and Supplementary Information).

We found that the best-fit model was able to predict very well all five empirical patterns (Fig. 2; R^2 from 0.42 to 0.73). Indeed, it captured the fact that the bulk of breeding migrants occur in the Northern Hemisphere around 50°N (Fig. 2a–d) and redistribute for the non-breeding season to the southern part of the Northern Hemisphere, with few species crossing the Equator (Fig. 2e–h). The model accurately predicted the peak in resident bird species in the tropics (Fig. 2i–l). It also captured a peculiarity in the global pattern of bird migration¹⁷. That is, the transition from avian communities that are net senders and those that are net receivers of breeding

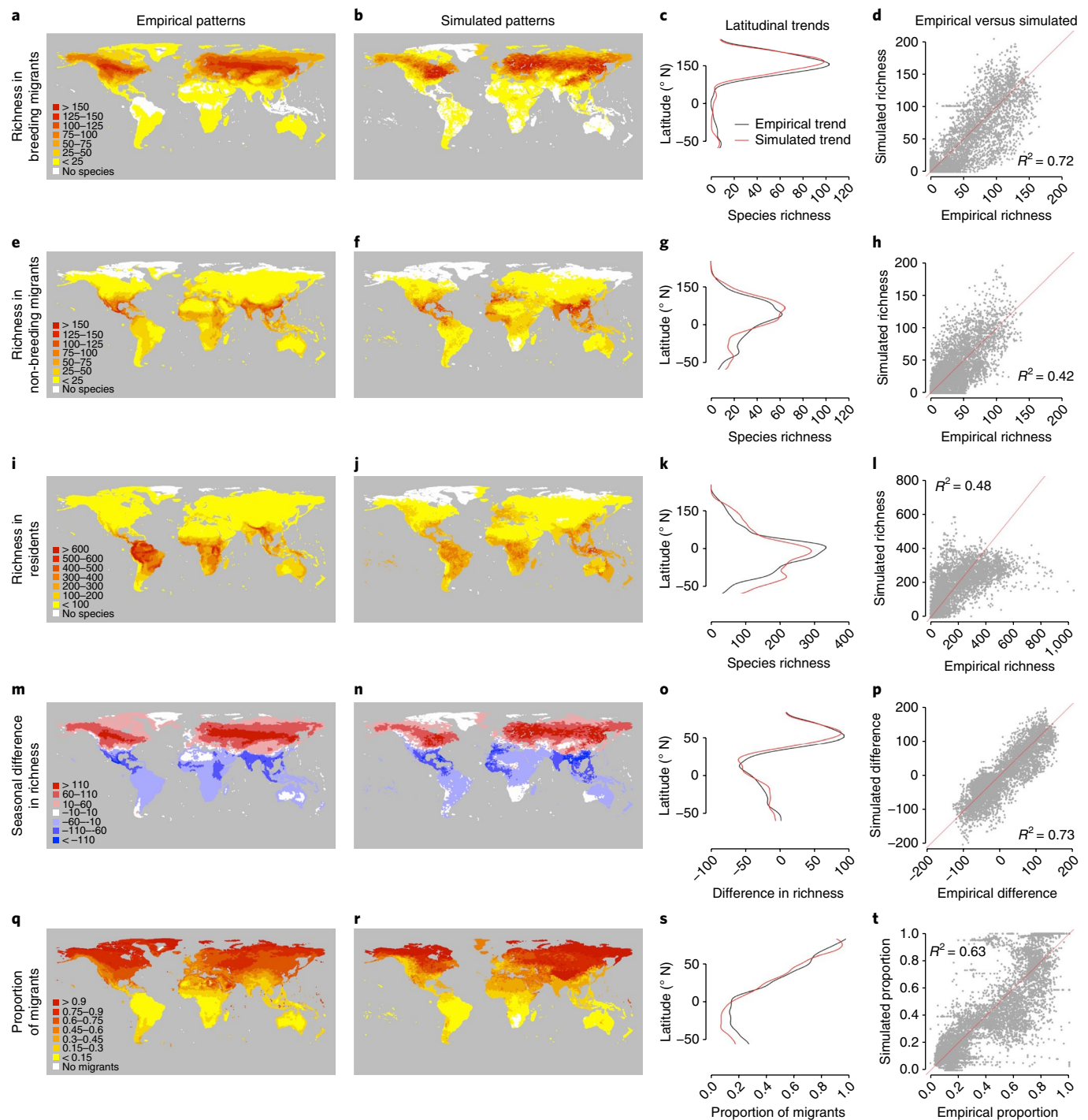


Fig. 2 | Contrast between empirical patterns in the global spatial distribution of terrestrial birds across seasons and the same patterns simulated through the overall best-fit model. a–d, Richness in breeding migrants. **e–h,** Richness in non-breeding migrants. **i–l,** Richness in residents. **m–p,** Seasonal difference in richness. **q–t,** Proportion of migrants. Latitudinal trends (**c,g,k,o,s**) were obtained using Nadaraya–Watson kernel regression estimates (using the *ksmooth* function from the *stats* package in R). In the scatterplots of the relationship between the empirical and simulated patterns (**d,h,l,p,t**), goodness-of-fit was computed using the sum of squared residuals from the 1:1 line (in red). In **r**, land hexagons with zero-simulated species (for which the proportion of migrants could not be calculated) are shaded in grey. A total of 7,783 virtual species were simulated.

migrants at around 35°N, a band of high species turnover with similar richness in both seasons (Fig. 2m–p). Finally, the model accurately predicted the latitudinal increase in the proportion of migrants from the Equator to the poles (Fig. 2q–t). Overall, the model captured the strong asymmetry between the Northern and Southern hemispheres very well when it came to patterns in bird diversity (Fig. 2)¹⁷.

To assess the significance of the predictive ability of our model, we contrasted it with a null model that did not take into account energetic constraints. The null model failed to predict any of the empirical patterns (Figs. 3 and 4 and Supplementary Fig. 3, all R^2 negative), and was significantly outperformed by the best-fit model (Fig. 4; $P < 0.001$). A null model in which energetic costs (that is, migration, thermoregulation and reproduction) were all set to zero

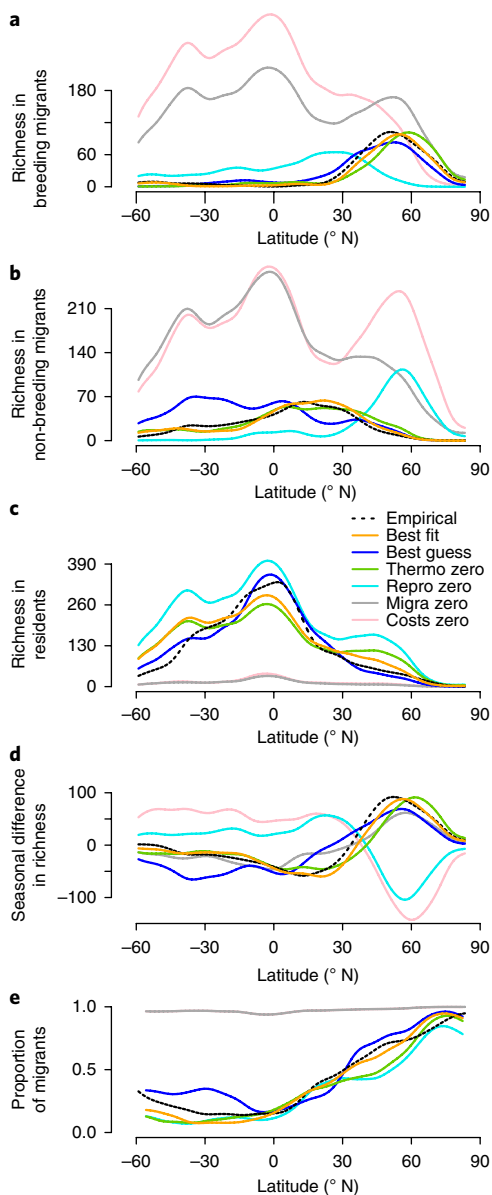


Fig. 3 | Latitudinal trends for empirical and simulated patterns in the global spatial distribution of terrestrial birds across seasons. **a**, Richness in breeding migrants. **b**, Richness in non-breeding migrants. **c**, Richness in residents. **d**, Seasonal difference in richness. **e**, Proportion of migrants. Latitudinal trends were obtained using Nadaraya-Watson kernel regression estimates (using the `ksmooth` function from the `stats` package in R). The legend in **c** applies to all panels.

(thus driven by energy availability only) also failed to produce realistic patterns (Figs. 3 and 4 and Supplementary Fig. 4, all R^2 negative). These results therefore confirm that energy efficiency (that is, the optimization of the balance between energy acquisition and energy expenditure) is the key mechanism underpinning the good predictive ability in our model, rather than simply energy supply. To assess the relative importance of the three components of energetic costs in driving the model outputs, we also tested null models whereby each of the components were removed in turn. When either migration or reproduction costs were set to zero, the model was no longer capable of producing realistic patterns (Figs. 3 and 4, and Supplementary Figs. 5 and 6), indicating that each of these components is crucial for understanding empirical patterns. In contrast, removing thermoregulation alone had relatively little effect on

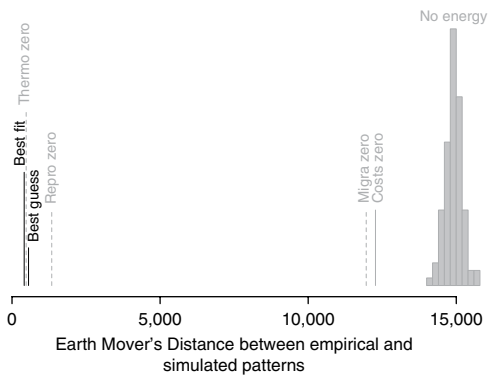


Fig. 4 | Predictive value of the overall best-fit model contrasted with the best-guess model and the null models. The best-guess model is based on parameters obtained from the literature. No energy refers to the null model that does not integrate energetic considerations (the species are selected randomly). Costs zero refers to the null model that does not account for energetic costs (the distributions are driven solely by energy supply). Migration (Migra) zero refers to the best-fit model for the model without migration cost. Thermoregulation (Thermo) zero refers to the best-fit model for the model without thermoregulation cost. Reproduction (Repro) zero refers to the best-fit model for the model without reproduction cost. The predictive value was quantified as mean Earth Mover's Distance values⁴⁹ between simulated and empirical patterns, whereby the smaller the distance the better the match.

model performance (Figs. 3 and 4 and Supplementary Fig. 7), suggesting this component is less important. This result could be due to the fact that cold climate is highly correlated with low resource availability. Therefore, thermoregulation does not add much to the explanatory power of the model for which output is already strongly driven by the geographical and seasonal variation in energy supply. In fact, including thermoregulation improves predictions in temperate areas with some relative mismatch between the winter normalized difference vegetation index (NDVI) (being relatively high) and winter temperature (being below the thermoneutral zone of species). Thermoregulation helps better predict the distribution of species during the winter in these temperate environments (for example, North America, Europe and East Asia) and, consequently, the latitudinal peak in breeding migrants (further southwards; compare Fig. 2 with Supplementary Fig. 7).

Further confirmation that our model realistically captures biological processes comes from the fact that the best-fit parameter values (obtained by fitting the full model to the empirical data) closely match the values estimated independently from the literature (Supplementary Fig. 8 and Supplementary Information). The best-guess model based on the latter set of parameter values also has excellent predictive ability, although not as well as the overall best-fit model (Fig. 4 and Supplementary Fig. 9), and deviations between the two are plausible given the data (Supplementary Information). Removing thermoregulation cost from the model also leads to parameter estimations that are realistic, but not when migration or reproduction costs were set to zero (Supplementary Fig. 8).

Discussion

Overall, the strong ability of our mechanistic model to predict macroecological patterns of bird diversity suggests that it successfully integrates the main costs and benefits affecting the global geographical distribution of terrestrial bird species. Our results also provide strong support for the hypothesis that these distributions are driven by energy efficiency.

The mismatches between the empirical patterns and those predicted by the full best-fit model are informative. In particular, the

model tends to under-predict richness in migrants in open habitat regions (for example, the western half of North America, central Asia and East Africa) while over-predicting it in regions dominated by forest (for example, the eastern half of North America, central and eastern Europe, and Southeast Asia) (Supplementary Fig. 10). This mismatch may reflect the limitations of using the NDVI as a proxy for locally available energy across different types of habitat. The model also under-predicts richness in resident species in major mountainous regions, mainly the Andes and the Himalayas (Supplementary Fig. 10c), which probably reflects processes that are not taken into account in this study, such as the effect of habitat and topographical heterogeneity on speciation rate, range size and endemism^{18–21} in these regions. This under-prediction is also what is probably affecting the fact that the model generates fewer virtual species than the empirical total number of species (7,783 species for the full best-fit model, 7,844 for the best-guess model and 9,356 for the empirical model). This result is most likely to be due to the model not being able to capture the high number of small-ranged, endemic species in areas with high heterogeneity in habitat and topography.

Our analysis provides a unified mechanistic explanation of bird migration as a behaviour that allows highly mobile avian species to optimize their energy budget in the face of fluctuating resources and interspecific competition. Indeed, in the early parts of the simulation, virtual species are resident in the tropics (Supplementary Fig. 11a), which is consistent with Fretwell's prediction that a bird "in a totally noncompetitive atmosphere, might find itself a resident"²². Migration is a progressively more favourable strategy as interspecific competition reduces local energy supply, making it more energetically efficient to migrate to areas with unexploited seasonal surpluses of energy. In our increasingly crowded virtual world, species progressively start exploiting more extreme pockets of seasonally available energy supply, often migrating longer distances (Supplementary Fig. 11b). Our results confirm the important role of the competitive reduction of energy supply, which has been proposed to be a key driver of bird migration^{22–24}, in shaping the distribution of migrant and resident bird species across the world and over the year.

While we focused on terrestrial birds, we expect our model to be directly transferable to other highly mobile taxa, such as oceanic fish or cetaceans. Its application to taxa requiring specific habitats during their life cycle (for example, cliffs for seabirds and caves for bats) would require integrating the distribution of such resources when considering possible distributions for virtual species. The principles of the model are more broadly applicable even to low-mobility taxa, such as amphibians, or even non-mobile taxa, such as plants, by assuming migration is too costly or impossible, such that only resident distributions are energetically efficient. These species have other mechanisms for dealing with seasonality, such as dormancy or hibernation, which could be easily added to our model by assuming different energetic requirements in different seasons.

A mechanistic explanation of how species distribute across the world is key to improving our capacity for predicting how ecosystems will respond to global change. Effects of climate change on temperature and precipitation²⁵ are already affecting thermoregulation costs and the distribution of energy supply. Anthropogenic land-use changes are increasing competition in areas where humans appropriate larger proportions of the energy supply (for example, through agriculture and urbanization²⁶), but also increasing resource supply in others (for example, in landfills²⁷). Our mechanistic model provides a useful tool for making predictions of such effects on biodiversity under various scenarios of global change.

Methods

Empirical bird data. Spatial polygons representing the global distribution of 9,783 non-marine bird species were obtained from BirdLife International and

NatureServe²⁸. The data and their treatment are described in detail in a previous publication¹⁷. Briefly, range maps were converted into presences and absences on a global grid of hexagons that were equal in area (~23,322 km²) and shape²⁹, with 7,352 hexagons remaining after removing the ones that contained no land and for which not all environmental data were available. After removing species for which the seasonal geographical distributions did not overlap with any remaining land hexagons, 9,356 species remained for the analysis (species occurring in both the western hemisphere (WH < 30° W) and eastern hemisphere (EH > 30° W) were treated for analytical purposes as two species). Migratory species were defined as those whose breeding and non-breeding distributions did not completely overlap, totalling 1,403 species. From these data, we generated the following five spatial patterns that captured the global seasonal distribution of terrestrial birds: 'richness in breeding migrants', the number of species present in each hexagon only during their breeding season; 'richness in non-breeding migrants', the number of species present in each hexagon only during their non-breeding season; 'richness in residents', the number of bird species present in each hexagon year-round; 'seasonal difference in richness', the number of breeding migrants minus the number of non-breeding migrants; and 'proportion of migrants', the number of migrant bird species (both breeding and non-breeding visitors), divided by the total number of bird species (both residents and migrants).

Virtual worlds. The virtual worlds we simulated have the same geography and seasonality as the real world as follows: a similar landmass distribution, mapped onto the same hexagonal grid as above; two seasons, corresponding to the northern summer (April to September) and northern winter (October to March); similar climate, as measured by temperature and precipitation; and similar resources, as measured by the NDVI. Since no migratory flyways cross the Atlantic Ocean or the Pacific Ocean, simulations were run separately for the WH and for the EH. Temperature and precipitation were computed for each hexagon in each season as the average across the corresponding six months using data from WorldClim at resolution 5'×30'. Monthly estimates of the NDVI were obtained from NASA's (National Aeronautics and Space Administration's) Earth Observatory at a resolution of 0.1°×0.1°. The estimated values were pre-processed as in an earlier study¹⁶, summed across the six months of each season and averaged for the pixels in each hexagon to obtain local seasonal NDVI values.

Virtual bird species. Each virtual species inhabits a particular combination of a breeding range and a non-breeding range, either congruent (resident species) or different (migratory species). Each range is composed of multiple local populations (one per hexagon), and a single hexagon may have populations of several species. We assumed that all species are functionally equivalent (ignoring differences in traits and ecology) and that all populations of all species are the same in their basal energy use for existence (thus assuming energetic equivalence between bird species^{32–34}). We also assumed that individuals are homogeneously distributed within each species' range. We therefore modelled an average species as follows: a body mass (M) of 38.6 g (median among empirical values reported in a previous publication³⁵), and an average geographical range size of 131 hexagons if in the WH or 180 hexagons if in the EH (median values in our empirical dataset).

Model overview. The Seasonally Explicit Distributions Simulator (SEDS) model is a mechanistic model of the geographical distribution of bird species across the world and throughout the year. The SEDS model was built using first ecological and energetics principles, relating the energy supply available in the environment with the energy requirements of bird species. It is a cost–benefit model whose common currency is energy, built from the following three main components: a module estimating the costs associated with species' energy requirements; a module estimating the spatial and seasonal variation in energy supply; and a module simulating virtual species' geographical range options. Integrating these three components, the model is applied through a sequence of simulation steps whereby a virtual world is progressively filled with virtual species until it becomes saturated (Fig. 1). Our model is not an evolutionary model of competition; it does not take into account the potential mortality associated with the processes or the potential increases in reproductive output under favourable conditions. The model essentially assumes species' populations to be at demographic equilibrium.

Energetic costs. For each local bird population, the energetic requirements were composed of the basal energy use for existence, BE_{ex} , which was set to be 1 (arbitrary) unit, and additional costs associated with migration (m_c), thermoregulation (t_c) and reproduction (r_c), which were converted into arbitrary units of energy use (Supplementary Fig. 1).

The m_c corresponds to the energetic cost of, each year, travelling twice between the breeding and non-breeding ranges. We assumed the following parameters: m_c increases linearly with distance travelled (thus, $m_c = 0$ for resident species), and migration happens instantaneously at the end of each season (its cost added to the corresponding season to reflect the previous investment in fat reserves). For each season, m_c was computed as a function of the great circle distance, d_m , between the centroids of the breeding and non-breeding geographical ranges (average distance travelled by individual birds of the species assuming that they migrate using the shortest route).

Thus $m_C = \alpha \times d_m$, with α as a free parameter determining the energetic requirements for a bird to travel a unit of distance (Supplementary Fig. 1).

The t_C corresponds to the additional energetic cost of maintaining a relatively constant internal body temperature for endotherms such as birds. Empirical and theoretical studies have indicated that species have a thermoneutral zone, a range of environmental temperatures (between a lower critical temperature T_{LC} and the upper critical temperature T_{UC}) within which $t_C = 0$, and outside of which costs increase linearly^{36,37}. We ignored the increase in energy demand above T_{UC} , since it has been poorly quantified previously, and only 12 hexagons in the global grid (0.16%) had an average seasonal temperature above the mean T_{UC} (34.3 °C) of 167 species sampled across the bird phylogeny and the world³⁸. We therefore focused on t_C when the ambient temperature is below T_{LC} . The Scholander–Irving model of heat transfer³⁹ describes how body temperature is regulated by balancing the rates of heat production and heat loss. Its core equation for a resting (endotherm) animal with minimal heat loss (by maximizing insulation and optimizing body posture) is $T_b - T_{LC} = \text{BMR}/C$, where T_b is the body temperature (40 °C for birds), BMR is the basal metabolic rate in $\text{ml O}_2 \text{ h}^{-1}$, and C is the rate of heat loss or thermal conductance. The T_{LC} of the thermoneutral zone can therefore be obtained as $T_{LC} = 40 - \text{BMR}/C$. We assumed that if the ambient temperature (T_a , in °C) is below the T_{LC} , the cost of thermoregulation increases linearly with the temperature difference between T_{LC} and T_a such that $t_C = (C/\text{BMR}) \times (T_{LC} - T_a)$ (Supplementary Fig. 1). So, if $\beta = \text{BMR}/C$, the cost of thermoregulation for a virtual species in a focal season was computed as a function of the ambient temperature with a free parameter β , as follows:

$$t_C = \begin{cases} \frac{40 - T_a - \beta}{\beta}, & \text{if } T_a < 40 - \beta \\ 0, & \text{if } T_a \geq 40 - \beta \end{cases}$$

The r_C value corresponds to the extra energy requirements for the production of new organisms. We assumed that this cost is all incurred in the breeding season, modelling it as a fraction of the BE_{UB} , with free parameter γ , as: $r_C = \gamma \times \text{BE}_{UB}$ (Supplementary Fig. 1).

For any local bird population i of a given species at a given season, the energetic costs were computed as follows: $E_{C_{BR}} = 1 + m_C + t_C + r_C$ for the breeding season and $E_{C_{NB}} = 1 + m_C + t_C$ for the non-breeding season, determined by the three free parameters (α , β and γ). The total energetic cost for a species at a given season was then computed as the mean of the energetic costs of the p local populations across the seasonal ranges. That is, $E_{C_{BR}} = (\sum_{i=1}^p E_{C_{BR}})/p$ for the breeding season and $E_{C_{NB}} = (\sum_{i=1}^p E_{C_{NB}})/p$ for the non-breeding season.

Energy supply. In each hexagon, energy supply, E_s , is the total amount of resources that is used to fuel bird species' metabolism. We modelled E_s as proportional to the NDVI, a remote-sensing measure of greenness that correlates well with primary productivity (often used in macroecological studies analysing the effect of the availability of energy and resources on the distribution of bird diversity^{14–16,40–42}). The NDVI varies across space (hexagons) and between seasons. We assumed that in any given hexagon j , the higher the NDVI the higher its carrying capacity for birds, such that $E_{sj} = \mu \times \text{NDVI}_j$, with μ a fourth free parameter of the model. This parameter was used for adjusting the energy supply (that is, acting as carrying capacity) for the model to generate a realistic total number of virtual species (that is, similar to the empirical number). The energy available, E_{Aj} , is equal to the energy supply minus the energy already consumed by species previously simulated and occurring in hexagon j . The energy available to a species at a given season was computed as the mean of the energy available in the p hexagons across the seasonal range. That is, $E_{A_{BR}} = (\sum_{j=1}^p E_{A_{BR}})/p$ for the breeding season and $E_{A_{NB}} = (\sum_{j=1}^p E_{A_{NB}})/p$ for the non-breeding season.

Virtual species' range options. We generated 1,000 contiguous geographical ranges in our virtual world (400 in the WH, 600 in the EH, reflecting differences in area) to serve as options from which the distributions of virtual species were simulated (Fig. 1). Ranges were generated using a method adapted from the spreading dye algorithm^{18,43} through a climate-driven approach of range expansion that has been shown to accurately capture the empirical distribution of bird ranges' shape⁴⁴. Each range was seeded from a single randomly selected hexagon and then allowed to stochastically spread into adjacent unoccupied hexagons, constrained by climatic conditions, until reaching a fixed size. For each range, an initial climatic optimum was obtained from the position of the seed hexagon in a climatic space defined by a mean annual temperature (z -standardized) and a mean annual precipitation (z -standardized after being log-transformed). We then selected two neighbours of the seed hexagon, with the probability of selection being higher for neighbours closer to the climatic optimum (that is, lower Euclidian distance d in the climatic space between itself and the climate optimum), calculated as $2(d+1)^{-30}$, divided by the sum of these values across all neighbours (hence decaying exponentially with increasing climatic distance d). We then repeated this procedure, each time redefining the climatic optimum as the average climatic condition across the already selected (that is, occupied) hexagons and selecting 25% (rounded to the larger integer) of the set of unoccupied neighbours of the occupied hexagons (summing the

probabilities of the ones being neighbours of more than one occupied hexagon), until the desired range size was reached (131 hexagons, ~ 3.05 million km^2 , in the WH, 180 hexagons, ~ 4.20 million km^2 , in the EH).

Simulation steps. Each simulation was performed on the WH and EH separately. At the start of the simulation (T_0), the virtual world is empty of bird species and the E_A in each hexagon is equal to the E_s in each season (Fig. 1). Then, each simulation step consisted of the following four sub-steps:

1. Generate candidate virtual species by combining all possible pairs of range options, each option occupied in either the breeding season or in the non-breeding season, and each species breeding in either the northern summer or the northern winter. There are therefore 320,000 candidate virtual species (from 400 range options) in the WH and 720,000 (from 600 range options) in the EH. This sub-step only needs to be performed once, and the same set of candidate virtual species is used in all steps of the simulation.
2. For each candidate distribution, compute a year-round energy-efficiency score (E) defined as energetic cost divided by energy available, combined for both seasons, such that $E = \frac{E_{C_{BR}}}{E_{A_{BR}}} + \frac{E_{C_{NB}}}{E_{A_{NB}}}$. The costs for each candidate species depend on its spatial location. That is, thermoregulation costs are higher in cooler regions and migration costs relate to the distance between breeding and non-breeding ranges. Reproduction costs are allocated only to the breeding range. Energy available also varies with spatial location, depending on the initial energy supply (the NDVI) as well as how much of it has been appropriated by previously simulated species.
3. The virtual species selected is the one with the lowest E (the most energy efficient).
4. The newly simulated virtual species consumes energy within its seasonal geographical ranges equivalent to its corresponding energetic cost, effectively depleting the energy available in all the hexagons across its geographical distribution as follows: $E_{A_{BR}} - E_{C_{BR}}$ across the breeding range and $E_{A_{NB}} - E_{C_{NB}}$ across the non-breeding range.

The simulation then proceeds, each step going through sub-steps 2, 3 and 4 to add a new virtual species.

The SEDS model therefore assumes that the scarcer the energy supply and the higher the number of other species competing for it, the harder (and therefore less energy efficient) it is to access it. Biologically, this result may correspond to costs associated with mutual interference^{45,46}, increasing search time⁴⁷ and territorial defence⁴⁸. As energy efficiency is higher when the difference between energetic costs and energy available is higher, it may be more energy efficient to occupy a geographical area with low resources and low competition than a crowded highly productive area. For example, taking a species with an average energetic cost of 1.5 units across its geographical range, if no other species are present, a geographical range in rainforest (average energy supply of 200, leading to an energy efficiency score of 0.0075) is more energy efficient than a range in desert (average energy supply of 30, leading to an energy efficiency score of 0.0500). But if the rainforest is already crowded with other (previously simulated) species using 90% of the energy supply (available energy supply of 20, leading to an energy efficiency score of 0.075), then the desert without competitors is a better option. Energy supply therefore acts as a carrying capacity that limits the number of populations of different species that can co-occur in any given hexagon. We stopped simulating species when 95% of the energy supply in at least one season across the world was used; that is, when the virtual world was nearly saturated with simulated species.

Parameter fitting the best-fit model. The SEDS model contains the following four free parameters: α , β , γ and μ . For any combination of parameter values, a virtual world saturated with virtual species can be deterministically simulated, from which it is possible to map each of the above-described global spatial patterns. We searched the four-dimensional parameter space for the combination of parameter values that best fit the empirical data. That is, it produces a virtual world in which the three basic patterns—richness in breeding migrants, richness in non-breeding migrants and richness in residents (the other two patterns being derived from combinations of these)—best match the empirical patterns. We computed the Earth Mover's Distance⁴⁹ to quantify the similarity between simulated and empirical spatial patterns, which is a measure of goodness-of-fit that is also sensitive to the number of virtual species allowed in the virtual world, and a genetic algorithm to fit the parameters (see Supplementary Information for details; results presented in Fig. 2).

Parameter estimating the best-guess model. For each of the four free parameters, we also obtained coarse estimates from the literature, wholly independent from the dataset we used (see Supplementary Information for details; results in Supplementary Fig. 8). We contrasted these estimates with the parameters obtained from the fitting process to evaluate the biological realism of the latter.

Null models. We tested whether our model based on energy efficiency produced significantly better results than expected by comparing its results with simulations performed using the following null models (Figs. 3 and 4):

A null model without integration of energetic considerations (called 'no energy'). As in the SEDS model, the simulation starts with a virtual world empty of species, but then the geographical distribution of the virtual species added in each step is selected at random among all the candidates, irrespective of energetic costs or energy supply. For each of 100 runs of this null model, we generated 3,886 virtual species for the WH and 5,206 virtual species for the EH, corresponding to the numbers in our empirical dataset.

A model in which the distribution of species is purely driven by energy supply, without taking into account costs (other than basal requirements for existence; called 'costs zero'). Here, we set costs of migration, thermoregulation and reproduction to 0 by using the following parameter values: $\alpha=0$, $\beta=80$, $\gamma=0$. As in the SEDS model, the simulation starts with a virtual world empty of species where the energy available is equal to the energy supply. The energetic requirements of each simulated species are simply the basal energy requirements for existence, set to 1. Each simulation step then consists of the four same sub-steps as in the SEDS model, also stopping when the virtual world is nearly saturated. The fitting procedure is used to estimate the value for parameter μ that best fit the data. A single best-fit simulation is obtained, as results are deterministic.

Three models in which a cost component (that is, m_c , t_c and r_c) is removed in turn. For the first of these models, called 'migra zero', the cost of migration is set to zero by using the following parameter value: $\alpha=0$. For the second of these models, called 'thermo zero', the cost of thermoregulation is set to zero by using the following parameter value: $\beta=80$. For the third of these models, called 'repro zero', the cost of reproduction is set to zero by using the following parameter value: $\gamma=0$. For each of these three models, simulations are run in the same way as the SEDS model, and the fitting procedure is used to estimate the value for the free parameters that best fit the data and a single best-fit simulation is obtained.

Reporting Summary. Further information on experimental design is available in the Nature Research Reporting Summary linked to this article.

Code availability. The computer code used for this study is available from the corresponding author upon request.

Data availability. The bird species distribution data are available for non-commercial use upon request to BirdLife International (<http://datazone.birdlife.org/species/requestdis>). The environmental data (NDVI and temperature) are freely available from the sources listed in the respective references.

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Author contributions

M.S. conceived the model and developed it with A.M. and A.S.L.R. M.S. also performed the analyses with support from A.M., and drafted the paper with conceptual and editorial input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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► Experimental design

1. Sample size

Describe how sample size was determined.

The analyses in this study are not based on sampling. Part of the results come from simulations, and part build from empirical data on the distribution of all 1403 terrestrial bird species that are migratory (i.e., "whose breeding and non-breeding distributions did not completely overlap" according to the spatial dataset in ref 28; see methods).

2. Data exclusions

Describe any data exclusions.

The spatial analyses are done on a hexagonal grid covering all of the Planet's land surface. The 7352 cells analysed exclude 924 hexagons that are not covered by one or more of the environmental datasets we used (which are derived from remote sensing data). This exclusion is not expected to affect the result, as these hexagons are mostly in polar regions (which have few or no terrestrial bird species) or remote islands (whereas our simulations consider contiguous ranges).

3. Replication

Describe whether the experimental findings were reliably reproduced.

As these analyses are not based on sampling, replication of results is not affected by the sample chosen. In the simulation part of the analyses, we generated a large number of simulated species (1,040,000) but for any given set of parameters the simulation is deterministic (arriving at a unique result). We repeated simulations for a wide range of parameters. The analysis of empirical data pertains to all 1403 migratory terrestrial bird species. As there is a single Planet Earth, replication of these analyses is not possible.

4. Randomization

Describe how samples/organisms/participants were allocated into experimental groups.

Not applicable to this study as the analyses were not based on samples.

5. Blinding

Describe whether the investigators were blinded to group allocation during data collection and/or analysis.

Not applicable to this study as the analyses were not based on samples.

Note: all studies involving animals and/or human research participants must disclose whether blinding and randomization were used.

6. Statistical parameters

For all figures and tables that use statistical methods, confirm that the following items are present in relevant figure legends (or in the Methods section if additional space is needed).

n/a Confirmed

- ☒ ☐ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement (animals, litters, cultures, etc.)
- ☒ ☐ A description of how samples were collected, noting whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☒ ☐ A statement indicating how many times each experiment was replicated
- ☒ ☐ The statistical test(s) used and whether they are one- or two-sided (note: only common tests should be described solely by name; more complex techniques should be described in the Methods section)
- ☒ ☐ A description of any assumptions or corrections, such as an adjustment for multiple comparisons
- ☒ ☐ The test results (e.g. *P* values) given as exact values whenever possible and with confidence intervals noted
- ☒ ☐ A clear description of statistics including central tendency (e.g. median, mean) and variation (e.g. standard deviation, interquartile range)
- ☒ ☐ Clearly defined error bars

See the web collection on [statistics for biologists](#) for further resources and guidance.

► Software

Policy information about [availability of computer code](#)

7. Software

Describe the software used to analyze the data in this study.

The model presented in this study was written in Python 2.7, and the figures and other data analysis were conducted using R version 3.4.3. The genetic algorithm used to obtain the best fit models was run on the OpenMOLE workflow engine (www.openmole.org).

For manuscripts utilizing custom algorithms or software that are central to the paper but not yet described in the published literature, software must be made available to editors and reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). *Nature Methods* [guidance for providing algorithms and software for publication](#) provides further information on this topic.

► Materials and reagents

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8. Materials availability

Indicate whether there are restrictions on availability of unique materials or if these materials are only available for distribution by a for-profit company.

The bird species distribution data are available for non-commercial use upon request to BirdLife International (<http://datazone.birdlife.org/species/requestdis>). The environmental data (NDVI, temperature) are freely available from the sources listed in the respective references.

9. Antibodies

Describe the antibodies used and how they were validated for use in the system under study (i.e. assay and species).

Non-applicable.

10. Eukaryotic cell lines

a. State the source of each eukaryotic cell line used.

Non-applicable.

b. Describe the method of cell line authentication used.

Non-applicable.

c. Report whether the cell lines were tested for mycoplasma contamination.

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11. Description of research animals

Provide details on animals and/or animal-derived materials used in the study.

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12. Description of human research participants

Describe the covariate-relevant population characteristics of the human research participants.

Analyses build from already existing data on the distribution of bird species. No animals were handled for the purposes of this study.

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